Richard Brooks

Postdoctoral Associate at Boston University
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Research interests: Galaxy Formation and Evolution | The Milky Way | Dark Matter

Professional Appointments

Postdoctoral Associate, Boston University, USA **Oct 2026 –**
Pre-Doctoral Fellow, Flatiron Institute, New York, USA **Jan 2024 – Jun 2024**

Education

PhD in Astrophysics, University College London, UK **Sept 2022 – Aug 2026**
Supervisors: Dr. Jason Sanders & Prof. Ofer Lahav
Title: Dynamical Disequilibrium in the Milky Way – Large Magellanic Cloud System
Funding: The Royal Society and University College London

Master of Physics and Astronomy (MPhys), Durham University, UK **2018 – 2022**
Supervisors: Prof. Carlos Frenk & Dr. Kyle Oman
Title: A Path to Revealing the Nature of Dark Matter by Neutral Gas

Research Internships

Summer Research Student, ICC, Durham University, UK **Jun 2022 – Sept 2022**
Summer Research Student, IoA, University of Cambridge, UK **Jun 2021 – Aug 2021**

Publications Summary

9 publications, 7 as lead author, 2 as co-author (full list below, SciX ORCID page)

Academic Service

- **Founder/coordinator** of *The Local Group* monthly and semester meetings, bringing together Boston-area researchers working on topics related to the Local Group, with a particular focus on dynamics, 2026 – present.
- **Founder/coordinator** of a weekly presentation series at University College London bringing together all levels of academic seniority across many research areas. This research-enabling environment encourages new collaborations and is prioritised towards early career researchers, 2024 – 2026.
- **Organiser** of the STFC Introduction to Astronomy Summer School 2025 held at University College London. Organised and chaired a panel discussion on 'The PhD experience'.
- **Reviewer** for the Astrophysical Journal (ApJ).

Research Mentorship

- **Rutong Pei**, Boston University graduate student, Velocity dipoles in FIRE, 2026 – present.
- **Teresa Su**, University College London PhD student, SBI for stellar ages, 2025 – present.
- **Peiyu Tsai**, University College London MSci student, time-dependent MW–LMC dark matter halo masses, 2025 – 2026.

Teaching

- **Guest Lecturer**, Galactic Astrophysics graduate class, CAS AS 791, Boston University, 2026.
- **Teaching Assistant**, University College London, 2023 – 2025:
  - Modern Physics, Astronomy and Cosmology
  - Atoms, Stars and the Universe
  - Physical Cosmology

Outreach

- **Science communicator** on social media, 2026 – present:
  - YouTube, Instagram and TikTok: *100k+ total views, 600+ total followers.*
  - Provides a realistic view of a career path in astrophysics.
  - Focused on science of the Milky Way and the wider Universe.
- JWST **outreach** event at the Science Museum, London, UK, 2023.

Publications List

Lead author papers

- *The Milky Way – Large Magellanic Cloud Interaction with Simulation Based Inference*
  Brooks, R. A. N; Sanders, J. L; Chandra, V; Garavito-Camargo, N; Dillamore, A; Price-Whelan, A; Ting, Yuan-Sen – (MNRAS, 2026)
- *Quantifying the Milky Way, LMC and their interaction using all-sky kinematics of outer halo stars*
  Brooks, R. A. N; Sanders, J. L; Dillamore, A; Chandra, V; Garavito-Camargo, N; Price-Whelan, A; Cargile, P – (MNRAS, 2026)
- *A Simulation Based Inference Approach to the Dynamics of the MW–LMC System – Validation*
  Brooks, R. A. N; Sanders, J. L; Dillamore, A; Garavito-Camargo, N; Price-Whelan, A – (MNRAS, 2026)
- *LMC Calls, Milky Way Halo Answers: Disentangling the Effects of the MW–LMC Interaction on Stellar Stream Populations*
  Brooks, R. A. N; Garavito-Camargo, N; Johnston, K. V; Price-Whelan, A; Sanders, J. L; Lilleengen, S. L. – (ApJ, 2025)
- *Action and energy clustering of stellar streams in deforming Milky Way dark matter haloes*
  Brooks, R. A. N; Sanders, J. L; Lilleengen, S. L; Petersen, M; Pontzen, A – (MNRAS, 2024)
- *The north-south asymmetry of the ALFALFA H*I *velocity width function*
  Brooks, R. A. N; Oman, K. A; Frenk, C. S – (MNRAS, 2023)

Co-authored papers

- *The velocity field of our Milky Way outer stellar halo based on DESI DR2*
  Li, S *et al.* incl. **Brooks, R. A. N** – (ApJ, 2026)
- *GSE vs. LMC: reshaping of radially biased stellar haloes by satellites*
  Dillamore, A; Sanders, J. L; **Brooks, R. A. N** – (MNRAS, 2026)

Conference Proceedings

- *Origins of the north-south asymmetry in the ALFALFA H*I *velocity width function*
  Brooks, R. A. N; Oman, K. A; Frenk, C. S – (IAU Symposium 379, 2023)

Talks

16 invited talks and seminars; 10 conference talks

Invited Talks & Seminars

- Departmental seminar, Lund University, Sweden — Apr 2026
- Departmental seminar, Niels Bohr Institute, Denmark — Apr 2026
- Friday Late Afternoon Talk, University College London, UK — Mar 2026
- Departmental seminar, University of Surrey, UK — Mar 2026
- Galaxies group meeting, Columbia University, USA — Oct 2025
- ITC Luncheon Talk, Center for Astrophysics, Harvard University, USA — Oct 2025
- CDT in Data Intensive Science, University College London, UK — May 2025
- MAT talk, Massachusetts Institute of Technology, USA — May 2024
- Conroy group meeting, Center for Astrophysics, Harvard University, USA — Mar 2024
- Hernquist group meeting, Center for Astrophysics, Harvard University, USA — Mar 2024
- Coffee talk, Institute for Advanced Study, Princeton, USA — Mar 2024
- Sanderson group meeting, University of Pennsylvania, USA — Mar 2024
- Milky Way Stars group meeting, Columbia University, USA — Mar 2024
- Surrey London Astro Meeting, University of Surrey, UK — Dec 2023
- Institute for Computational Cosmology, Durham University, UK — Apr 2023
- Seminar, University of Helsinki, Finland — Mar 2023

Conference Talks

- SBI for Galaxy Evolution Conference, University of Cambridge, UK — Jun 2026
- IAU 403: The Hidden Beauty of the Galactic Outskirts, Córdoba, Argentina — Oct 2025
- UK National Astronomy Meeting, Durham University, UK (two talks) — Jul 2025
- Milky Clouds over Yellowstone, Bozeman, Montana, USA — May 2025
- Streams24 Workshop, Durham University, UK — Aug 2024
- Small Galaxies, Cosmic Questions II, Durham University, UK (flash talk) — Jul 2024
- CCA Pre-Doc Symposium, Flatiron Institute, USA — Jun 2024
- Milky Clouds over Manhattan, Flatiron Institute, USA — Feb 2024
- IAUS 379: Dynamical Masses of Local Group Galaxies, Potsdam, Germany — Mar 2023

Collaborations

- Rubin Observatory – Dark Energy Science Collaboration; member of the dark matter and stellar streams working groups.
- Spec-S5 – member of the Near Field Cosmology working group.